WORKED SOLUTION

STEP II 1993, Question 3

Pure • Riccati Differential Equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q3

The key idea

Both equations are nonlinear because of the y^2 term, so none of the standard first-order methods applies directly. The substitutions are designed to kill that y^2 term. In (i), writing y as a multiple of u'/u makes the squared terms cancel and leaves a linear second-order equation with constant coefficients. In (ii), $y = 1/x$ happens to be an exact solution, and writing $y = 1/x + 1/v$ turns the equation into a linear first-order equation for v, which an integrating factor solves.

Solution

Part (i)

Put $y = -\dfrac{u'}{3u}$, where $u' = \dfrac{\mathrm{d}u}{\mathrm{d}x}$. By the quotient rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{u''}{3u} + \frac{(u')^2}{3u^2}, \qquad \text{and} \qquad 3y^2 = 3 \cdot \frac{(u')^2}{9u^2} = \frac{(u')^2}{3u^2}.$$

So the awkward terms cancel:

$$\frac{\mathrm{d}y}{\mathrm{d}x} - 3y^2 = -\frac{u''}{3u}.$$

The equation $\dfrac{\mathrm{d}y}{\mathrm{d}x} - y - 3y^2 = -2$ therefore becomes

$$-\frac{u''}{3u} + \frac{u'}{3u} = -2.$$

Multiplying by $-3u$ (where $u \neq 0$, which is needed anyway for y to be defined) gives

$$u'' - u' - 6u = 0.$$

The auxiliary equation is $m^2 - m - 6 = (m-3)(m+2) = 0$, so $m = 3$ or $m = -2$, and

$$u = Ae^{3x} + Be^{-2x}$$

for constants A and B, not both zero. Then $u' = 3Ae^{3x} - 2Be^{-2x}$ and

$$y = -\frac{u'}{3u} = \frac{2Be^{-2x} - 3Ae^{3x}}{3\left(Ae^{3x} + Be^{-2x}\right)}.$$

Ask yourself A first-order equation should have one arbitrary constant. Why do we have two?

Only the ratio of A and B matters: multiplying u by a non-zero constant doesn't change u'/u. If $A \neq 0$, divide top and bottom by Ae^{3x} and write $K = B/A$:

$$y = \frac{2Ke^{-5x} - 3}{3\left(1 + Ke^{-5x}\right)} = \frac{2K - 3e^{5x}}{3\left(e^{5x} + K\right)},$$

where K is an arbitrary constant. The case $A = 0$ gives the constant solution $y = \frac{2}{3}$, which is the limit of this family as $K \to \infty$. Taking $K = 0$ gives the other constant solution, $y = -1$. As a check, the constant solutions satisfy $-y - 3y^2 = -2$, that is, $(3y - 2)(y + 1) = 0$.

Part (ii)

Put $y = \frac{1}{x} + \frac{1}{v}$. Then

$$\frac{dy}{dx} = -\frac{1}{x^2} - \frac{v'}{v^2},$$

and so

$$\begin{aligned} x^2\frac{dy}{dx} &= -1 - \frac{x^2v'}{v^2}, \\ xy &= 1 + \frac{x}{v}, \\ x^2y^2 &= x^2\left(\frac{1}{x^2} + \frac{2}{xv} + \frac{1}{v^2}\right) = 1 + \frac{2x}{v} + \frac{x^2}{v^2}. \end{aligned}$$

Adding these three lines, the equation $x^2y' + xy + x^2y^2 = 1$ becomes

$$1 - \frac{x^2v'}{v^2} + \frac{3x}{v} + \frac{x^2}{v^2} = 1, \qquad \text{that is,} \qquad -\frac{x^2v'}{v^2} + \frac{3x}{v} + \frac{x^2}{v^2} = 0.$$

Multiplying by v^2/x^2 (with $x \neq 0$, as the equation requires) gives the linear equation

$$\frac{dv}{dx} - \frac{3}{x}v = 1.$$

The integrating factor is $\exp\left(-\int \frac{3}{x}\,dx\right) = x^{-3}$ (on either of the intervals $x > 0$ or $x < 0$; we can drop the modulus sign because a constant multiple of an integrating factor works just as well). Hence

$$\frac{d}{dx}\left(\frac{v}{x^3}\right) = \frac{1}{x^3} \quad \Longrightarrow \quad \frac{v}{x^3} = -\frac{1}{2x^2} + c \quad \Longrightarrow \quad v = cx^3 - \tfrac{1}{2}x = \frac{x\left(Kx^2 - 1\right)}{2},$$

where c is an arbitrary constant and $K = 2c$. Therefore

$$y = \frac{1}{x} + \frac{2}{x\left(Kx^2 - 1\right)} = \frac{Kx^2 + 1}{x\left(Kx^2 - 1\right)},$$

with K an arbitrary constant. Taking $K = 0$ gives $y = -1/x$. The substitution assumed $1/v$ exists, so it misses $y = 1/x$ itself ("$v = \infty$"). This is also a solution, since $x^2\left(-\frac{1}{x^2}\right) + 1 + 1 = 1$, and it is the limit of the family as $K \to \infty$.

Checks

- In (i), at $x = 0$ the solution is $y(0) = \dfrac{2K - 3}{3(1 + K)}$. As $x \to +\infty$, $y \to -1$, and as $x \to -\infty$, $y \to \frac{2}{3}$ (when $K \neq 0$). So solutions run from one constant solution to the other, which is what we'd expect.
- In (ii), take $K = 1$ and $x = 2$: $y = \frac{5}{6}$ and $y'(2) = -\frac{31}{36}$. Then $x^2y' + xy + x^2y^2 = -\frac{31}{9} + \frac{5}{3} + \frac{25}{9} = 1$, as required.
- We checked both general solutions numerically, by finite differences, at hundreds of random points and values of K. The residual in the original equation was zero to within rounding error.

Common slips

- Leaving two constants A, B in the answer to (i) without noticing that only their ratio matters.
- Forgetting the factor 2 in the cross term of x^2y^2 in (ii), which turns $3v/x$ into $2v/x$ and gives the wrong integrating factor.
- Losing the solution $y = 1/x$ (or $y = \frac{2}{3}$ in (i)) that the substitution can't reach.

Beyond STEP

Equations of the form $y' = p(x) + q(x)y + r(x)y^2$ are called Riccati equations. Both tricks here are general. The substitution $y = -u'/(ru)$ always turns a Riccati equation into a linear second-order equation. And if you know one particular solution y_1, then $y = y_1 + 1/v$ always gives a linear first-order equation for v. This link between Riccati equations and linear second-order equations comes up again in quantum mechanics and control theory. Boyce and DiPrima, *Elementary Differential Equations*, has exercises on Riccati equations.

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